



Synthetic Dental X-Ray Generation and Segmentation Analysis

Valerio Spagnoli - 1973484
Computer Vision - Engineering in Computer Science
Sapienza University of Rome





Table of contents

1

Problem
Statement

3

Proposed Method

5

Model Evaluation

2

State of the Art

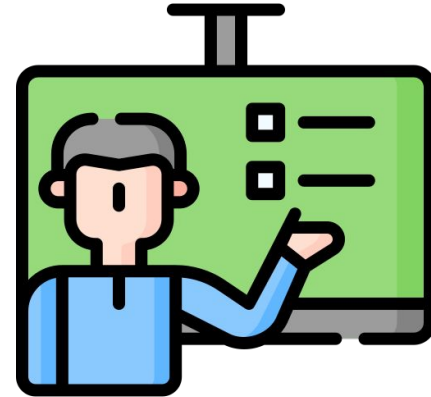
4

Dataset and
Experimental Setup

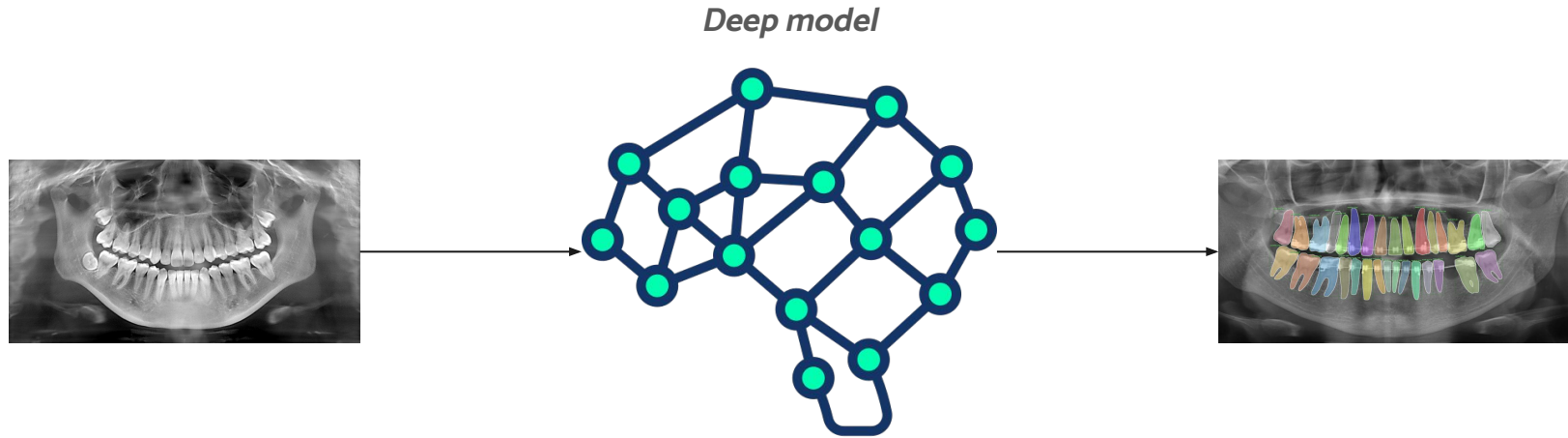


1 | Problem Statement

Why we need synthetic medical images - The ***Data Scarcity*** problem



Problem Statement





Problem Statement

Few annotations

There are few annotated images available for models training

Privacy constraints

X-Ray data collection is limited by privacy constraints

Limited generalization

Available datasets show limited diversity



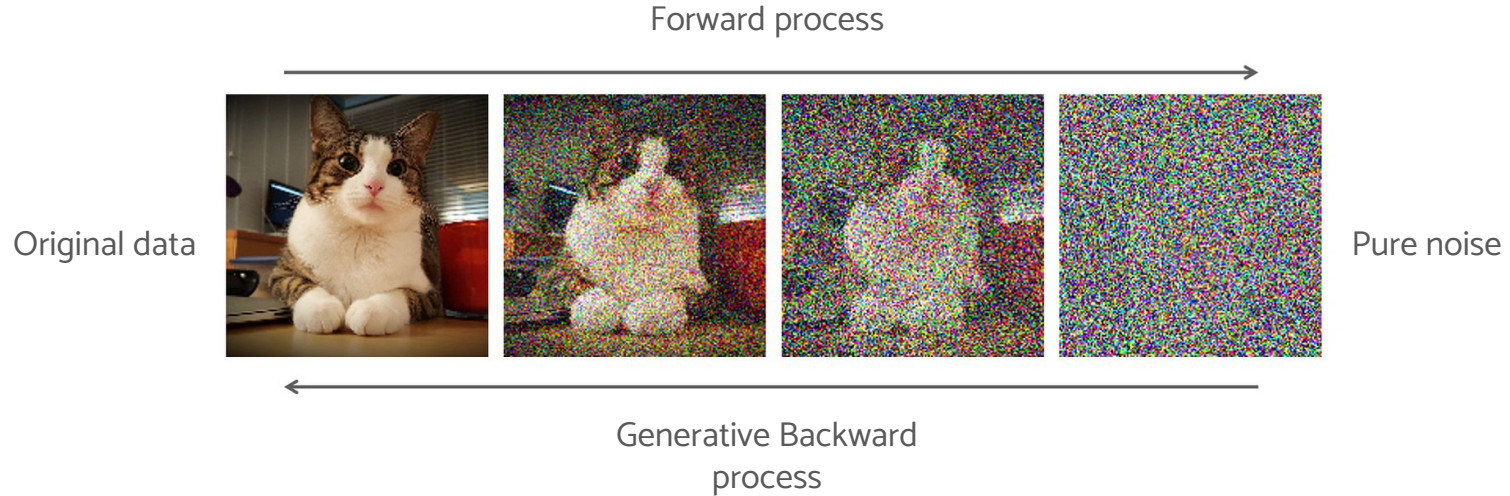
2 | State of the Art

Denoising Diffusion Probabilistic
Models - DDPM



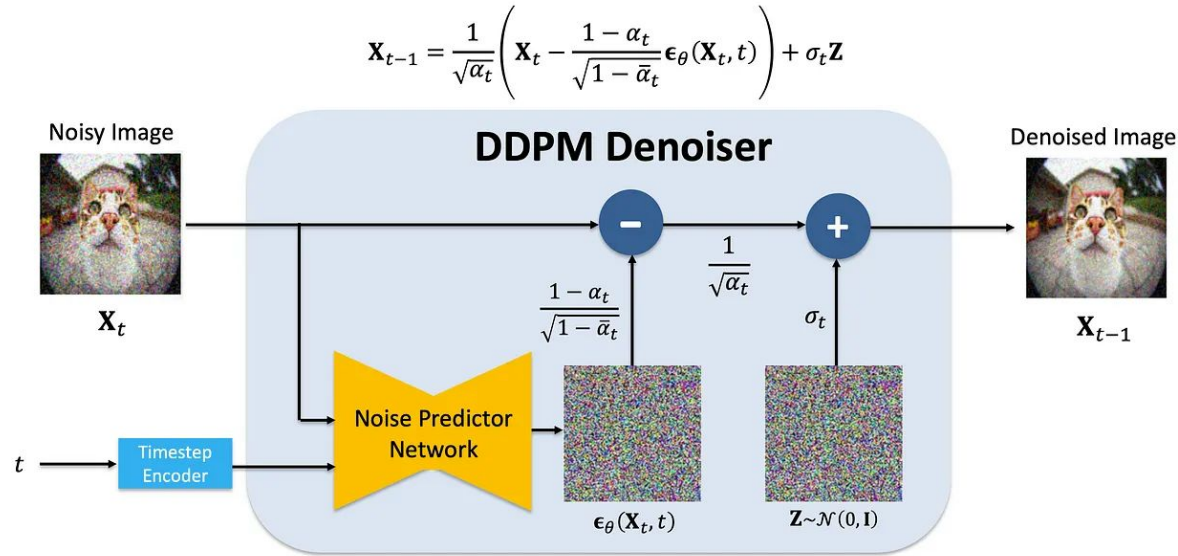
State of the Art

DIFFUSION MODELS - DDPM



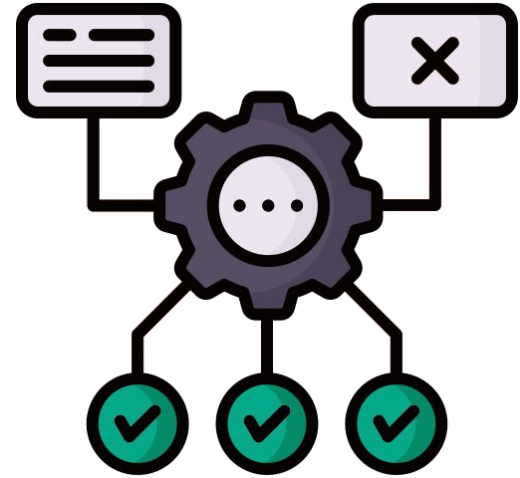
State of the Art

DIFFUSION MODELS - DDPM outline



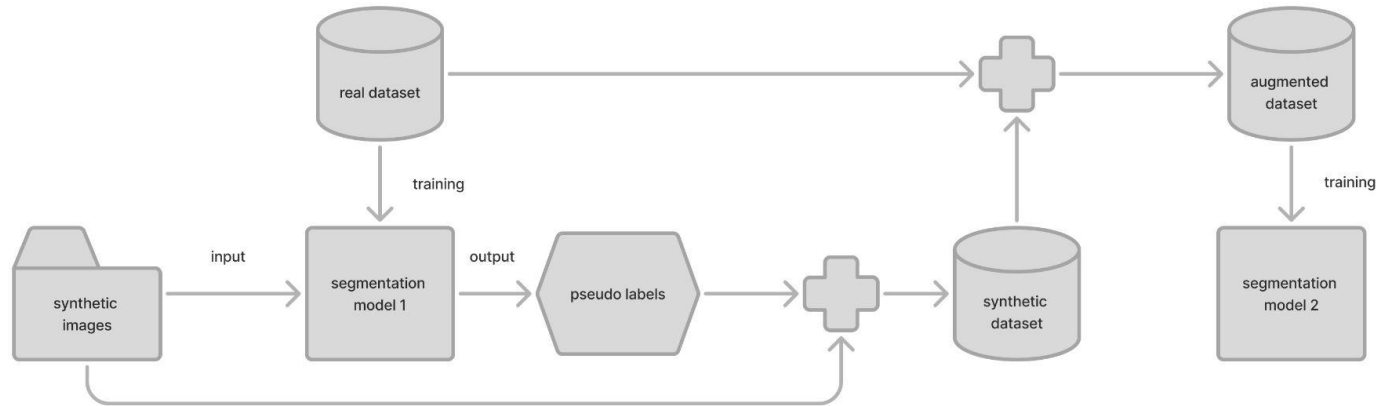
3 | Proposed Method

WGAN-GP and DDPM



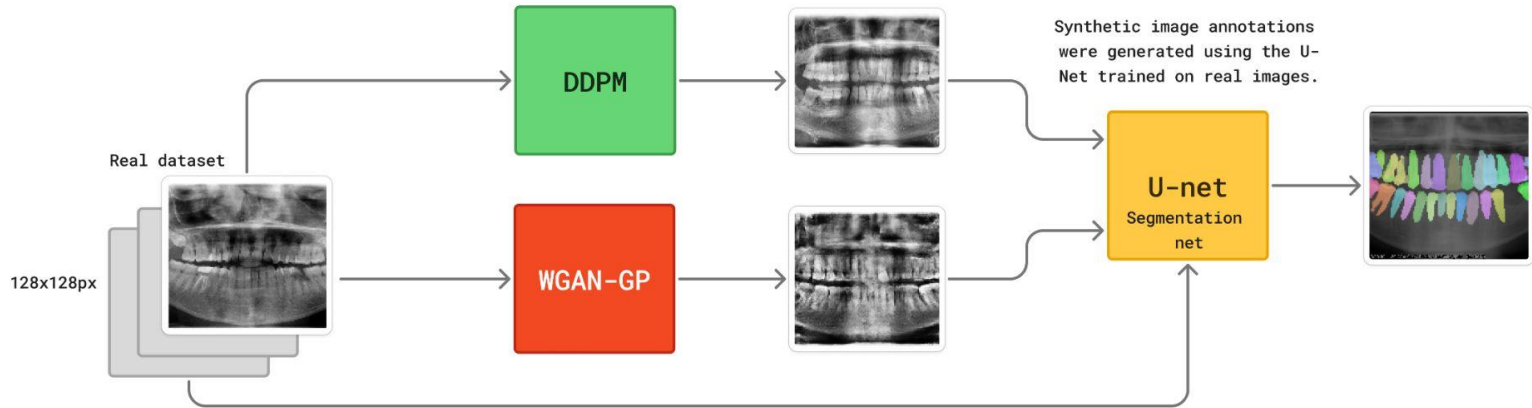
Proposed Method

Core idea to improve the segmentation model



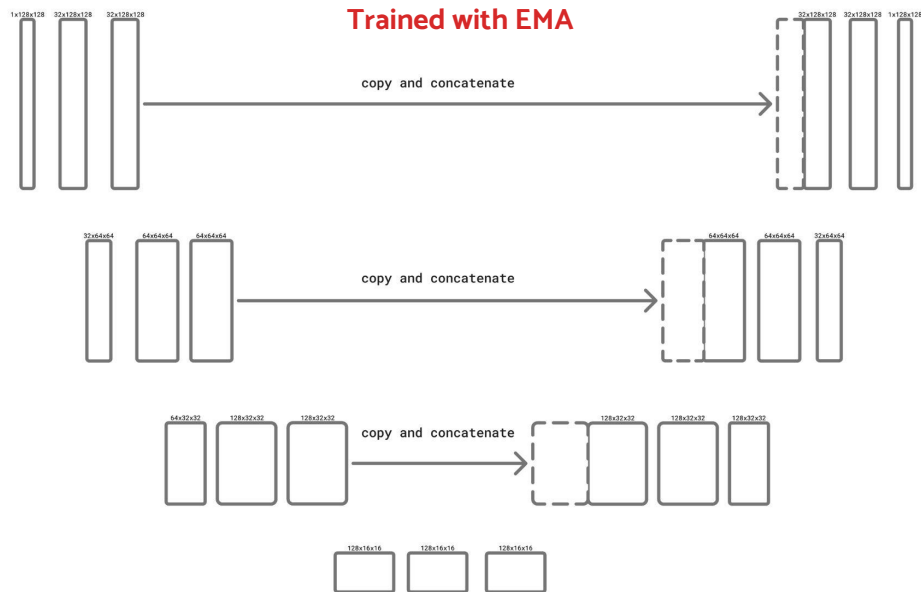
Proposed Method

Pipeline



Proposed Method

Diffusion model backbone



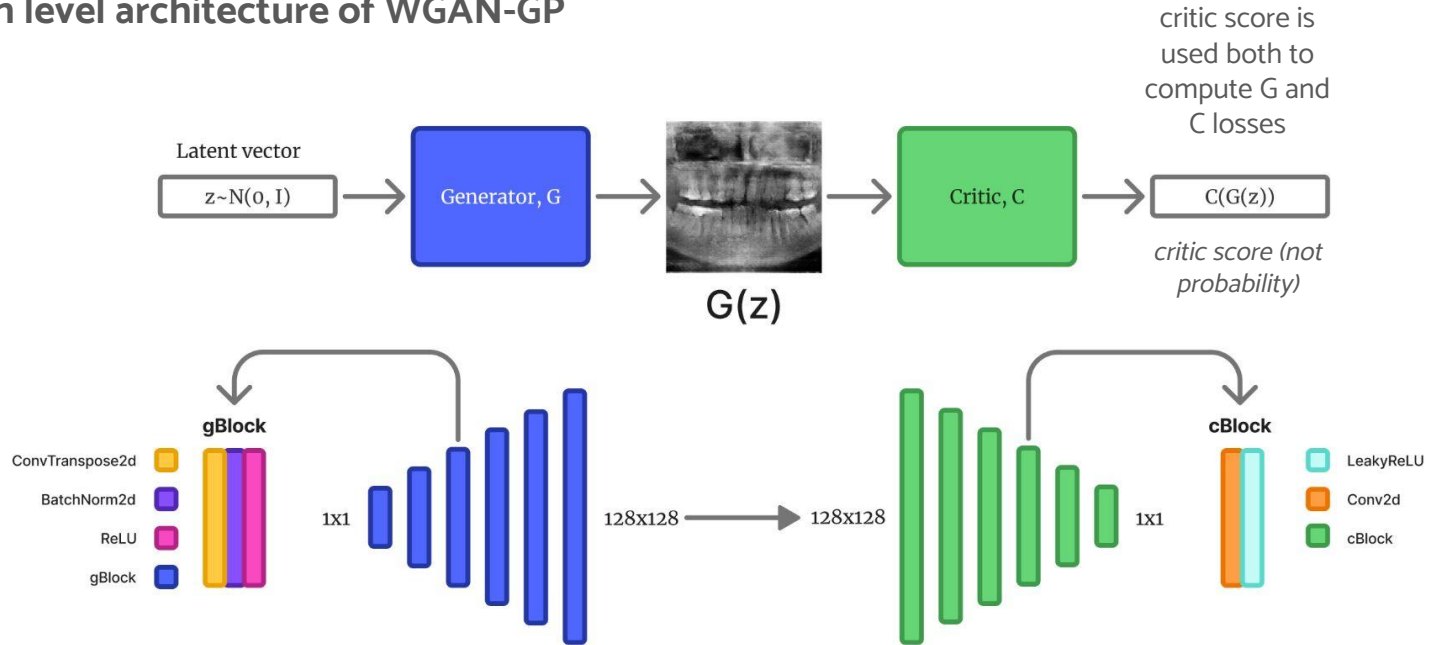
In the adopted ddpm, this U-Net is extended with:

- Residual blocks
- Self attention blocks
- Time embedding integration

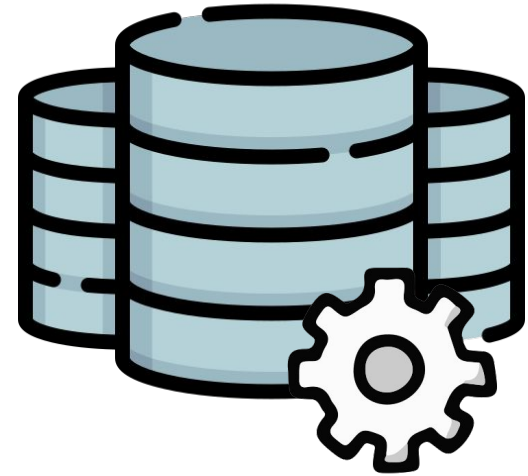


Proposed Method

High level architecture of WGAN-GP



4 | Dataset and Experimental setup



Dataset



HUMANS IN THE LOOP · UPDATED 3 YEARS AGO

46

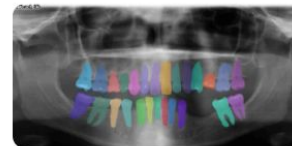
<> Code

Download



Teeth Segmentation on dental X-ray images

The dataset consists of 598 images with a total of 15,318 polygons



DENTEX

Dental Enumeration and Diagnosis
on Panoramic X-rays Challenge



Dataset

Generation
models

```
data/
├── train/ # hybrid (DENTEX + kaggle)
└── true/
```

Seg
model

```
data/
├── training_set/
│   ├── xrays/ # kaggle
│   ├── masks/
│   ├── diff_xrays/ # only diff generated
│   ├── diff_masks/
│   ├── mix_diff_xrays/ # real + diff
│   ├── mix_diff_masks/
│   ├── mix_wgan_xrays/ # real + wgan
│   └── mix_wgan_masks/
└── validation_set/
    ├── xrays/
    └── masks/
```



Experimental Setup

WGAN-GP setup

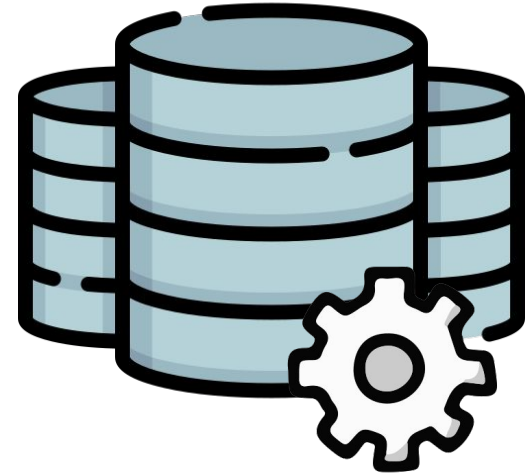
Hyperparameter	Value
Batch size	16
Latent dimension (z)	100
Generator channels	64
Critic channels	64
Generator LR	1×10^{-4}
Critic LR	5×10^{-5}
Gradient penalty λ	7
Optimizer	Adam
Adam (β_1, β_2)	(0.0, 0.9)
Loss	Wasserstein + GP

DDPM setup

Hyperparameter	Value
Noise schedule (β)	Linear (1e-4, 0.02)
Diffusion steps (T)	450
Batch size	16
Optimizer	Adam
Learning rate	1e-4
Loss	MSE
EMA learning rate	8e-5
EMA decay	0.999
Training steps	100,000

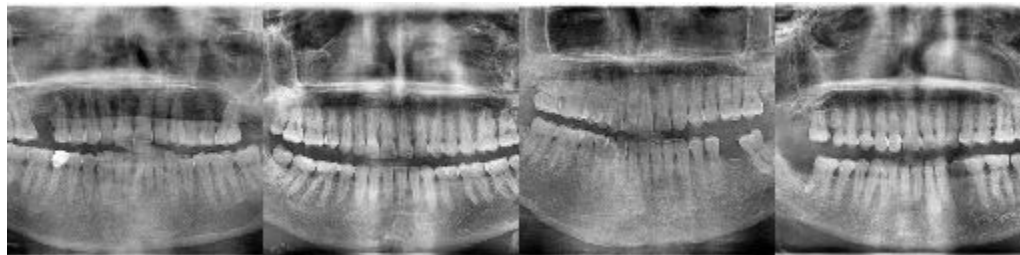


5 | Model Evaluation

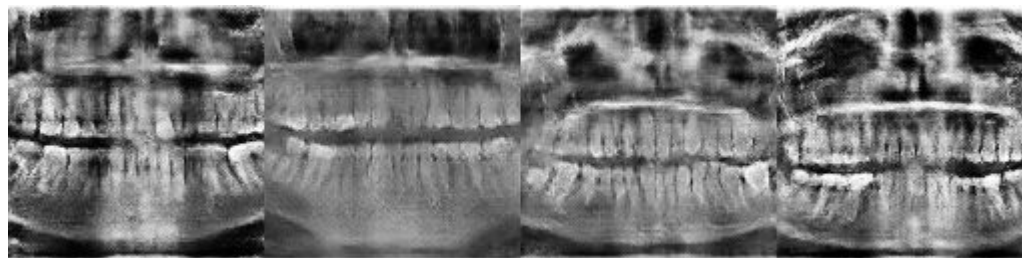


Model Evaluation

DDPM



WGAN-GP





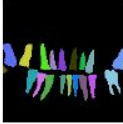
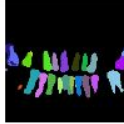
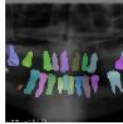
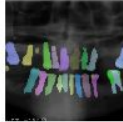
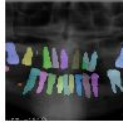
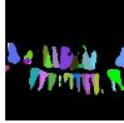
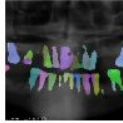
Model Evaluation

carried out from 1000 samples of
synthetic images

Model	FID	IS (real, fake, noise)
DDPM	55.3	(2.05, 1.66, 1.05)
WGAN-GP	151.7	(2.02, 1.70, 1.05)



Model Evaluation

Model	Image	GT	Pred	Overlay	mIoU	mF1	Recall	Precision	Accuracy
Legacy					0.5953	0.7294	0.7209	0.7718	0.9071
Mix DDPM					0.6020	0.7388	0.7355	0.7646	0.9063
Mix WGAN-GP					0.5853	0.7224	0.7173	0.7550	0.9035
Full DDPM					0.2516	0.3775	0.3580	0.4546	0.8122



References

- **Papers:**
 - AI Diffusion Models Generate Realistic Synthetic Dental Radiographs Using a Limited Dataset, Brian Kirkwood, Byeong Yeob Choi, James Bynum and Jose Slinas.
 - Pano-GAN: A Depp Generative Model for Panoramic Dental Radiographs, Soren Pedersen, Sanyman Jain, Mikkel Chavez, Viktor Ladehoff, Bruna Neves de Freitas and Ruben Pauwels.
- **Slides:**
 - [pietro-nardelli/sapienza-ppt-template](https://pietro-nardelli.sapienza-ppt-template)
[Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0)]
- **EMA code:** <https://medium.com/@heyamit10/exponential-moving-average-ema-in-pytorch-eb8b6f1718eb>





Thank you for the attention!

